

Due: Tuesday, April 15th, before class.

Name:

Collaborators:

Document your work and note any collaborators. In your write-up, please neatly document any code you have written, alongside its output, and including any necessary commentary to answer the question. Please use an `.ipynb` notebook for any code, include commentary in markdown code blocks, and print the `.ipynb` notebook to PDF to attach to your submission. The L^AT_EXsource for this file may be helpful: <https://www.overleaf.com/read/rrrfjtxxtrqf#763d6b>.

Please make sure to label the parts of your assignment! Use markdown blocks in your Julia notebook and make headings using `#`, `##`, and `###`.

Consider an image that consists of a rectangle of 200 by 300 pixels. This is a total of 60,000 pixels. If the image is a standard grayscale image, each pixel will be assigned an integer value between 0 (representing black) and 255 (representing white). The image file represents a large 200×300 matrix with integer values in this range.

Images – at least ones we would want to look at – have a lot of structure to them. Because of the structure, the information in them can often be conveyed very accurately by specifying far fewer numbers than the values for each and every pixel. Here is one way to do this: Let $\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$ be a singular value decomposition of an $m \times n$ $\mathbf{A}$. As usual, put

$$\mathbf{V} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_r] \quad \text{and} \quad \mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_r].$$

For any integer k with $0 < k < r$, we have

$$\mathbf{A}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top.$$

Although $\mathbf{A}_k$ is an $m \times n$ matrix, just like $\mathbf{A}$, the data needed to specify $\mathbf{A}_k$ is just the data needed to specify the k vectors in $\mathbb{R}^m$, the k vectors in $\mathbb{R}^n$, and the k numbers $\{\sigma_i\}_{i=1}^k$. That is, we need $k(1 + m + n)$ values to store $\mathbf{A}_k$, rather than mn values to $\mathbf{A}$. This can be *much* less if m and n are large.

Bounds similar to the ones we see in class show that for every pair of indices i, j , we have

$$|\mathbf{A}_{i,j} - (\mathbf{A}_k)_{i,j}| < \sigma_{k+1}.$$

Therefore, if we choose k large enough that $\sigma_{k+1} < 1$, the entries of $\mathbf{A}_k$ will all be within one grayscale value of the entries of $\mathbf{A}$, and so the images that the two matrices represent will be very close. Even if σ_{k+1} is not *so* small, this will often be the case.

In any case, the fact that the matrix $\mathbf{A}$ comes from a visually interesting image, and is not just a random matrix, gives it interesting structure that results in a rapid drop off in size of the singular values. Most of the information in an image matrix is contained in the few largest singular values, and their corresponding left and right singular vectors. This project explores these ideas.

Goals:

- (a) Write a function that imports an image file and convert it into a matrix (see Lectures 20 - 22). This function should take in the image file name and return a matrix. Note that if you work in color, you will need one matrix per channel (red, green, and blue).
- (b) Write a function that takes a given $m \times n$ matrix with floating point entries, and converts it to an image file. Be sure you round and truncate the values properly!
- (c) Write a function that takes as input an integer k , a $m \times n$ matrix $\mathbf{A}$, and returns $\mathbf{A}_k$, σ_1 , and σ_{k+1} .
- (d) Choose two images and apply your functions. Convert the images to a matrix, find the best rank- k approximation for $k = 2, 5, 10, 50$. Convert these back into images, and print them out, together with the original image. Beside each image, give the values of σ_1 and σ_{k+1} .
- (e) For these two images, identify for what value of k we have $\sigma_{k+1} < 1$. Find the best rank- k approximation for this value of k and convert this back to an image. Print this image beside the original images.
- (f) Try this compression with an image of you. How small can you make k and still keep the portrait recognizable? In other words, what is *your* rank?